

CHEMISTRY 3AB

MULTIPLE CHOICE QUESTIONS BOOKLET

Time allowed for this paper

Reading time before commencing work:	ten (10) minutes
Working time for paper:	three (3) hours

Materials required/recommended for this paper

To be provided by the supervisor

Booklet 1 – Multiple choice questions booklet

Booklet 2 - Question/Answer booklet with multiple choice answer sheet.

Chemistry Data sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid/tape, ruler, highlighters

Special items: non-programmable calculators satisfying the conditions set by the School Curriculum and Standards Authority for this course

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks Available	Percentage of exam
Section One: Multiple-choice	25	25	30	50	25
Section Two: Short answer	10	10	70	70	35
Section Three: Extended answer	4	4	80	80	40
Total				200	100

Instructions to candidates

1. Write your name and your teacher's name on **Booklet 2 only**.
2. The rules for the conduct of Western Australian external examinations are detailed in the *Student Information Handbook 2013*. Sitting this examination implies that you agree to abide by these rules.
3. Answer the questions according to the following instructions.

Section One: Answer all questions on the Multiple-choice answer sheet attached to the front of Booklet 2. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Sections Two and Three: Write your answers in Booklet 2.

4. When calculating numerical answers, show your working or reasoning clearly unless instructed otherwise.
5. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
6. Spare pages are included at the end of Booklet 2. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages in planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number.Fill in the number of question(s) that you are continuing to answer at the top of the page.

See next page

Section One: Multiple-choice

25% (50 marks)

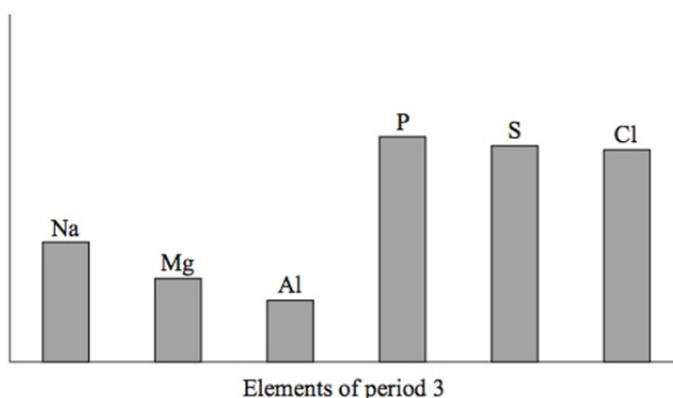
This section has **25** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 30 minutes.

1. What is the number of protons, electrons and neutrons in the $^{81}_{35}\text{Br}^{5+}$ ion?

- | | | | |
|-----|------------|--------------|-------------|
| (a) | 35 protons | 40 electrons | 46 neutrons |
| (b) | 35 protons | 30 electrons | 46 neutrons |
| (c) | 35 protons | 30 electrons | 81 neutrons |
| (d) | 35 protons | 40 electrons | 81 neutrons |

2. The horizontal axis of the bar chart below represents the elements of period 3 from sodium to chlorine (excluding silicon). What could the vertical axis represent?



- (a) Melting point of the element
- (b) Electronegativity of the bonded atom
- (c) Ionic radius of the most common ion
- (d) First ionization energy in the gaseous state
3. Which of the following covalent bonds would be expected to have the **greatest** polarity?
- (a) N – F
- (b) P – F
- (c) S – F
- (d) O – F

See next page

Questions 4 and 5 are based upon the table below, which gives the first eight ionisation energies (in kJ mol^{-1}) of four elements, all of which are in the first 20 elements of the Periodic Table.

Element	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
(i)	590	1,145	4,912	6,491	8,153	10,496	12,270	14,206
(ii)	577	1,820	2,740	11,600	14,800	18,400	22,400	27,500
(iii)	495	4,562	6,910	9,543	13,354	16,613	20,117	25,496
(iv)	1000	2260	3390	4560	6990	8490	27,200	37,100

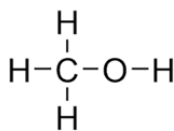
4. Which element would you expect to have the highest melting point?
- (a) (i)
(b) (ii)
(c) (iii)
(d) (iv)
5. Which element would you expect to have the lowest electronegativity?
- (a) (i)
(b) (ii)
(c) (iii)
(d) (iv)
6. Which of the following structural features is common to both diamond and graphite?
- (a) Covalent bonds between carbon atoms
(b) Delocalised electrons
(c) Dipole-dipole interaction
(d) Covalent dipole attraction
7. Which of the following properties do **not** have an increasing trend across Period 3 of the Periodic Table?
- (i) Atomic number
(ii) Atomic radius
(iii) Electronegativity
(iv) Ionization energy
(v) Number of electron shells
- (a) (i) and (iv)
(b) (i), (iii) and (iv)
(c) (ii) and (v)
(d) (i), (iii), (iv) and (v)

See next page

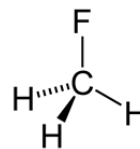
8. In which of the following liquids would you expect hydrogen bonding to be present between their molecules



Hydrogen fluoride



Methanol



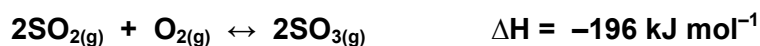
Fluoromethane

- (a) all of them.
- (b) hydrogen fluoride and methanol only.
- (c) hydrogen fluoride only
- (d) methanol only
9. Which of the following **best** explains why water has a lower boiling point than sodium chloride?
- (a) The covalent bonds in water are weaker than the ionic bonds in sodium chloride.
- (b) The intermolecular forces between water molecules are weaker than ionic bonds in sodium chloride.
- (c) The molecular mass of water (18.016) is less than the molecular mass of sodium chloride (58.44).
- (d) One mole of water occupies a smaller volume (18.02 mL) than one mole of sodium chloride at the same temperature and atmospheric pressure (27.01 mL).
10. The reaction between sulfur dioxide gas and oxygen gas to form sulfur trioxide gas is part of the synthesis of sulfuric acid by the Contact Process. It is catalysed by vanadium (V) oxide and after a certain time, equilibrium in the closed system is reached. Consider the following statements about the role of the catalyst in this reaction.
- (i) The catalyst allows the reaction to reach equilibrium sooner by increasing the rate of both the forward and reverse reactions.
- (ii) The catalyst lowers the activation energy barrier for both the forward and reverse reactions.
- (iii) The catalyst increases the mass of products compared to the mass of reactants when equilibrium is reached.

Which of the following statements is correct?

- (a) Only statement (i) above is true.
- (b) All the above statements are true.
- (c) None of the above statements is true.
- (d) Only statements (i) and (ii) above are true.

11. Consider the following reaction at equilibrium:

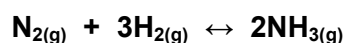


Which of the following would increase the rate of the **reverse** reaction, once equilibrium is re-established?

- (i) Adding more SO_2
- (ii) Adding more SO_3
- (iii) Increasing the temperature
- (iv) Decreasing the volume of the container

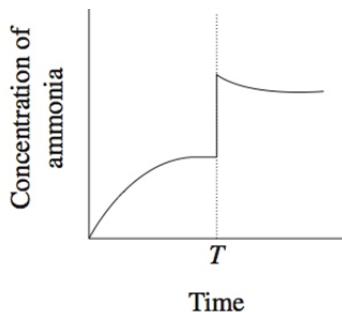
- (a) (i), (ii) and (iv)
- (b) (i), (iii) and (iv)
- (c) (ii), (iii) and (iv)
- (d) All of them

12. The following equation shows an equilibrium established in the synthesis of ammonia from its component gases:

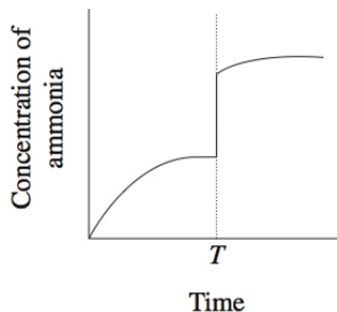


If the volume of the reaction chamber is suddenly halved (i.e. a reduction of 50%) at time T , which of the following graphs best depicts changes in the concentration of **ammonia** over time?

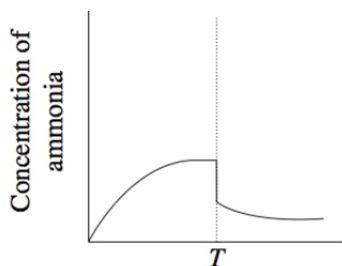
(a)



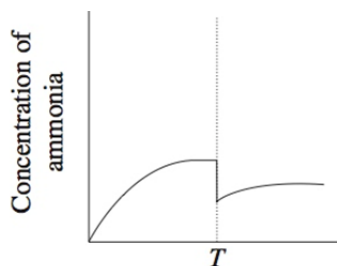
(b)



(c)



(d)

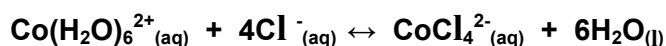


Questions 13, 14 and 15 refer to the following reaction between phosphorus trichloride and chlorine to form phosphorus pentachloride.



13. If phosphorus trichloride and chlorine were placed in a sealed and insulated container together with a catalyst, which of the following would **not** cause an increase in the rate at which equilibrium would be obtained?
- (a) Increasing the volume of the container
 - (b) Increasing the state of subdivision of the catalyst
 - (c) Increasing the temperature
 - (d) All of the above
14. Which of the following statements is/are **true** when the system is at equilibrium?
- i. Reactants are no longer turning into products
 - ii. The concentration of chlorine in the container stays unchanged.
 - iii. Adding a catalyst would not affect the proportions of the reaction mixture
- (a) (iii) only
 - (b) (i) and (ii) only
 - (c) (ii) and (iii) only
 - (d) (i), (ii) and (iii)
15. Which of the following changes would **not** be observed once equilibrium is re-established if a little phosphorus trichloride were added at constant temperature to the equilibrium mixture above?
- (a) The rate of the forward reaction would increase.
 - (b) The rate of the reverse reaction would increase.
 - (c) The concentration of phosphorus trichloride would increase.
 - (d) The value of the equilibrium constant (K) would decrease.

16. When chloride ions are added to a solution containing $\text{Co}(\text{H}_2\text{O})_6^{2+}$ the following equilibrium is established:



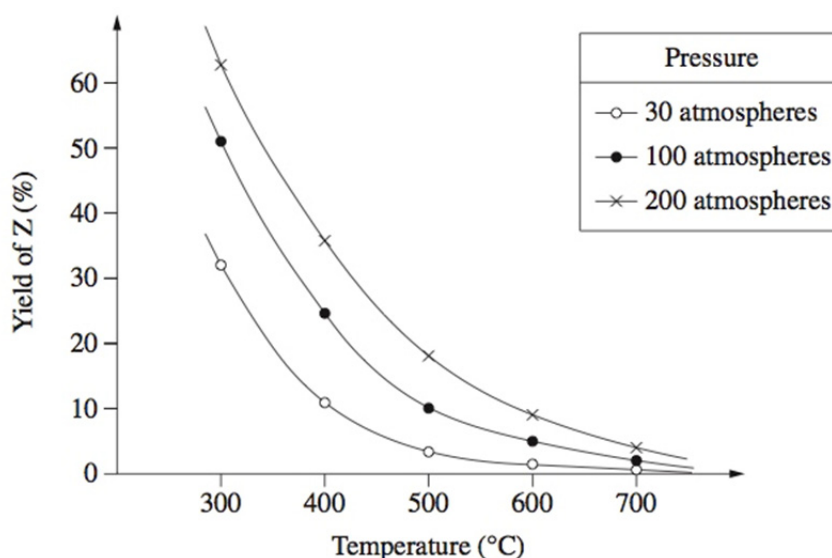
Pink

Blue

Solutions containing $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and Cl^- are frequently violet in colour owing to the presence of significant amounts of both $\text{Co}(\text{H}_2\text{O})_6^{2+}$ and CoCl_4^{2-} .

Which of the following statements concerning such solutions is **true**?

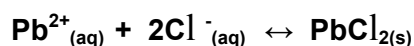
- (a) If the forward reaction is endothermic, cooling the solution will make the colour turn pink.
- (b) Adding a large amount of solid sodium chloride to the solution will make the colour turn pink.
- (c) Diluting the solution with water will make the colour turn blue.
- (d) If the forward reaction is exothermic, heating the solution will make the colour turn blue.
17. The graph below represents the yield of an equilibrium reaction at different temperature and pressure conditions inside a reaction vessel.



Which of the following reactions could produce the trends shown in the graph?

- (a) $\text{X}_{(\text{g})} + \text{Y}_{(\text{g})} \leftrightarrow 3\text{Z}_{(\text{g})}$ $\Delta H = +250\text{kJ}$
- (b) $\text{X}_{(\text{g})} + \text{Y}_{(\text{g})} \leftrightarrow 2\text{Z}_{(\text{g})}$ $\Delta H = -250\text{kJ}$
- (c) $2\text{X}_{(\text{g})} + 2\text{Y}_{(\text{g})} \leftrightarrow \text{Z}_{(\text{g})}$ $\Delta H = +250\text{kJ}$
- (d) $4\text{X}_{(\text{g})} + 2\text{Y}_{(\text{g})} \leftrightarrow 3\text{Z}_{(\text{g})}$ $\Delta H = -250\text{kJ}$

18. Consider a solution of lead (II) chloride at 25°C in equilibrium with some solid lead (II) chloride at the bottom of the solution.



If the forward reaction is exothermic, which of the following is the **best** way to increase the concentration of the lead and chloride ions in the solution?

- (a) Allowing some of the water to evaporate and then leaving the system to reach equilibrium again.
 - (b) Heating the system to 30°C.
 - (c) Adding some more solid lead (II) chloride.
 - (d) Sealing the container and vigorously shaking the solution.
19. Which one the following solutions has the highest concentration of dissolved ions at 25°C?
- (a) 0.4 molL⁻¹ CH₃COOH
 - (b) 0.3 molL⁻¹ Mg(NO₃)₂
 - (c) 0.5 molL⁻¹ Al (OH)₃
 - (d) 0.4molL⁻¹ Na₃PO₄
20. Four students were asked to test a solution for the presence of a cation by using various anions. The students obtained these results:

<i>Student</i>	<i>Chloride</i>	<i>Sulfate</i>	<i>Carbonate</i>
Acka-Dacka	No precipitate	No precipitate	Precipitate
Beeba	Precipitate	Precipitate	No precipitate
Cindy-Ella	Precipitate	Precipitate	Precipitate
Darth	No precipitate	Precipitate	No precipitate

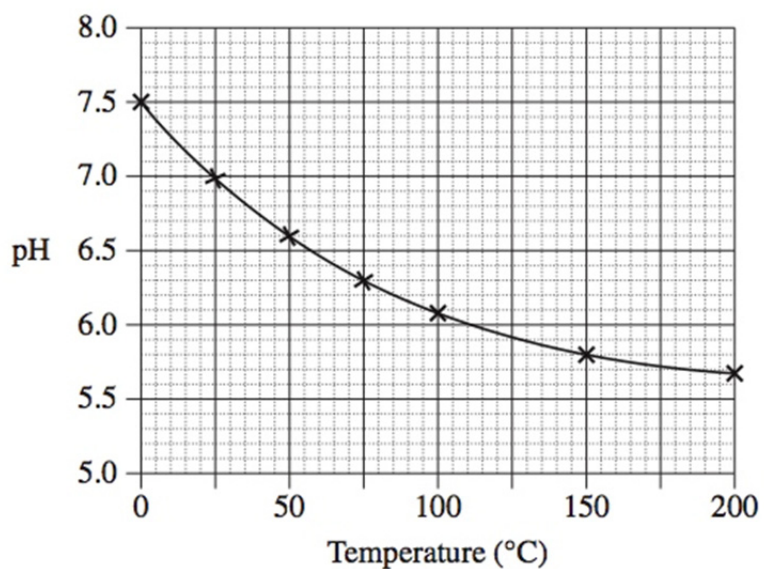
Each student concluded that Pb²⁺ was present.

Which student had results consistent with this conclusion?

- (a) Acka-Dacka
- (b) Beeba
- (c) Cindy-Ella
- (d) Darth

21. The following sets of solutions are mixed. In which of these will there be no visible reaction?
- (a) Set 1 consisting of barium hydroxide, dilute sulfuric acid, potassium nitrate.
 - (b) Set 2 consisting of potassium phosphate, copper(II) sulfate and sodium chloride.
 - (c) Set 3 consisting of sodium chloride, copper(II) sulfate and silver nitrate.
 - (d) Set 4 consisting of sodium iodide, barium nitrate and potassium hydroxide
22. The solubility of oxygen in seawater at 20°C is 14.4 mgL⁻¹. Which of the following is the concentration of oxygen in seawater in molL⁻¹?
- (a) 4.5 x 10⁻¹ mol L⁻¹
 - (b) 9.0 x 10⁻¹ mol L⁻¹
 - (c) 4.5 x 10⁻⁴ mol L⁻¹
 - (d) 9.0 x 10⁻⁴ mol L⁻¹
23. Which of the following correctly shows an acid- conjugate base pair?
- $$\text{HCO}_3^- (\text{aq}) + \text{H}_2\text{O} (\text{l}) \leftrightarrow \text{CO}_3^{2-} (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq})$$
- (a) HCO₃⁻ and H₃O⁺
 - (b) H₂O and CO₃²⁻
 - (c) HCO₃⁻ and CO₃²⁻
 - (d) All of the above
24. Which one of the following species is acting as a **Lowry-Bronsted acid** when ammonia gas is dissolved in a salt solution containing water and fully dissolved sodium chloride.
- (a) NH₃
 - (b) H₂O
 - (c) Na⁺
 - (d) Cl⁻

25. The graph below shows the pH of a solution of a weak acid, HA, as a function of temperature.



What happens as the temperature decreases?

- (a) HA becomes less ionised and the $\text{H}^+_{(\text{aq})}$ concentration increases.
- (b) HA becomes less ionised and the $\text{H}^+_{(\text{aq})}$ concentration decreases.
- (c) HA becomes more ionised and the $\text{H}^+_{(\text{aq})}$ concentration increases.
- (d) HA becomes more ionised and the $\text{H}^+_{(\text{aq})}$ concentration decreases.

END OF BOOKLET 1

Kira College
CHEMISTRY 12 SEMESTER 1 EXAMINATION 2015
Booklet 2



Name: _____ Teacher: _____

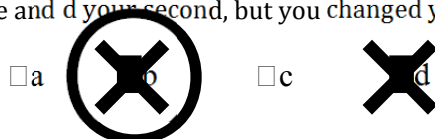
For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. For example, if b is your answer: ☐ a ☒ b ☐ c ☐ d

If you make a mistake, place a cross through that square, do not erase or use correction fluid. Shade your new answer.

For example, if b is a mistake and d is your correct answer: ☐ a ☒ b ☐ c ☒ d

In the event that you then change your mind back to your original answer, you then cross out the second selection and then circle the first choice.

For example, if b was the first choice and d your second, but you changed your mind back and b is your answer:



Multiple Choice: Mark the answer by shading in the box to the LEFT of the letter you chose.

- | | |
|---|---|
| 1. ☐ a ☐ b ☐ c ☐ d | 14. ☐ a ☐ b ☐ c ☐ d |
| 2. ☐ a ☐ b ☐ c ☐ d | 15. ☐ a ☐ b ☐ c ☐ d |
| 3. ☐ a ☐ b ☐ c ☐ d | 16. ☐ a ☐ b ☐ c ☐ d |
| 4. ☐ a ☐ b ☐ c ☐ d | 17. ☐ a ☐ b ☐ c ☐ d |
| 5. ☐ a ☐ b ☐ c ☐ d | 18. ☐ a ☐ b ☐ c ☐ d |
| 6. ☐ a ☐ b ☐ c ☐ d | 19. ☐ a ☐ b ☐ c ☐ d |
| 7. ☐ a ☐ b ☐ c ☐ d | 20. ☐ a ☐ b ☐ c ☐ d |
| 8. ☐ a ☐ b ☐ c ☐ d | 21. ☐ a ☐ b ☐ c ☐ d |
| 9. ☐ a ☐ b ☐ c ☐ d | 22. ☐ a ☐ b ☐ c ☐ d |
| 10. ☐ a ☐ b ☐ c ☐ d | 23. ☐ a ☐ b ☐ c ☐ d |
| 11. ☐ a ☐ b ☐ c ☐ d | 24. ☐ a ☐ b ☐ c ☐ d |
| 12. ☐ a ☐ b ☐ c ☐ d | 25. ☐ a ☐ b ☐ c ☐ d |
| 13. ☐ a ☐ b ☐ c ☐ d | |

Section Two: Short answer**35% (70 marks)**

This section has **10** questions. Answer **all** questions. Write your answers in the spaces provided.

When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to three (3) significant figures and include appropriate units where applicable.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

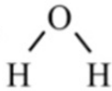
Suggested working time: 70 minutes

Question 26**(9 marks)**

Complete the table below by drawing the Lewis structures of the molecules listed and either drawing or naming the shape of the molecule.

All valence shell electron pairs should be represented either as : or as — .

The first row has been completed as an example.

Molecule	Lewis diagram	Sketch or name of shape	Number of <u>unbonded</u> electrons in molecule
H ₂ O	$\text{H}:\ddot{\text{O}}:\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$ or $\text{H}-\ddot{\text{O}}-\text{H}$	bent, or 	4 (or 2 pairs)
PH ₃			
CF ₄			
SO ₃			

See next page

(4 marks)

[illegible]

(10 marks)

(a) For each of these molecules, choose the description that best describes the polarity of the covalent bond and molecule. (2 marks)

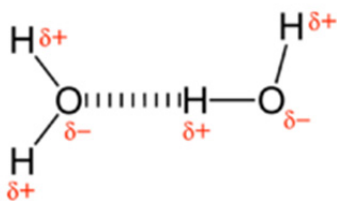
Molecule	Polarity of covalent bonds (write polar or non-polar)	Polarity of molecule (write polar or non-polar)
NH ₃		
SO ₃		

(b) For each of the following solids, state the **most dominant** form of intermolecular attraction and rank them in **increasing** order of melting point (i.e. 1 for lowest melting point and 4 for highest melting point). (4 marks)

Solid	Name of most dominant intermolecular force in solid state	Relative melting point ranking (write 1, 2, 3 or 4)
NH ₃		
PH ₃		
AsH ₃		
H ₂ O		

See next page

(c) The diagram below shows the physical attraction between neighbouring water molecules.



Sodium fluoride (NaF) and hydrogen fluoride are substances that are soluble in water.

i. Use the diagram above as a guide to show the **most dominant** form of physical attraction that exist between

- sodium fluoride and water molecules
- hydrogen fluoride and water molecules.

Your diagrams should include dotted lines (----) to show the relevant attraction as well as the location of charge densities (δ^+ and δ^-)

ii. Name the type of physical interaction in each case e.g. dispersion forces. (4 marks)

Sodium fluoride and water molecules	Hydrogen fluoride and water molecules
Name of most dominant attractive force	Name of most dominant attractive force

Question 29

(4 marks)

For the following reactions, describe fully the **observed changes**, including any:

- colour changes
- odours
- precipitates (give the colour)
- gases evolved (give the colour or describe as colourless).

(a) A sealed tube containing the equilibrium mixture, $2\text{NO}_{2(g)} \leftrightarrow \text{N}_2\text{O}_{4(g)}$ $\Delta H = -90 \text{ kJmol}^{-1}$ is placed in a beaker of hot water. (1 mark)

(b) Barium nitrate solution is added to potassium phosphate solution. (1 mark)

(c) Copper(II) chloride solid is added to sodium carbonate solution. (2 marks)

See next page

Question 30**(4 marks)**

Write balanced **ionic** equations to represent the reactions described below.

(a) Sodium hydroxide solution is added to iron(III) iodide solution.

(2 marks)

(b) Dilute hydrochloric acid is added to potassium chromate (K_2CrO_4) solution.

(2 marks)

Question 31**(6 marks)**

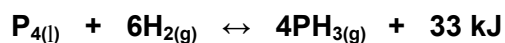
For each of the following pairs of substances, provide details of a **chemical** test that would allow the two substances to be distinguished from one another. Equations are **not** required.

Substances	Chemical test	Observations
$Na_2SO_{4(s)}$ and $NaCH_3COO_{(s)}$		with $Na_2SO_{4(s)}$: with $NaCH_3COO_{(s)}$:
$KOH_{(aq)}$ and $K_3PO_{4(aq)}$		with $KOH_{(aq)}$: with $K_3PO_{4(aq)}$:

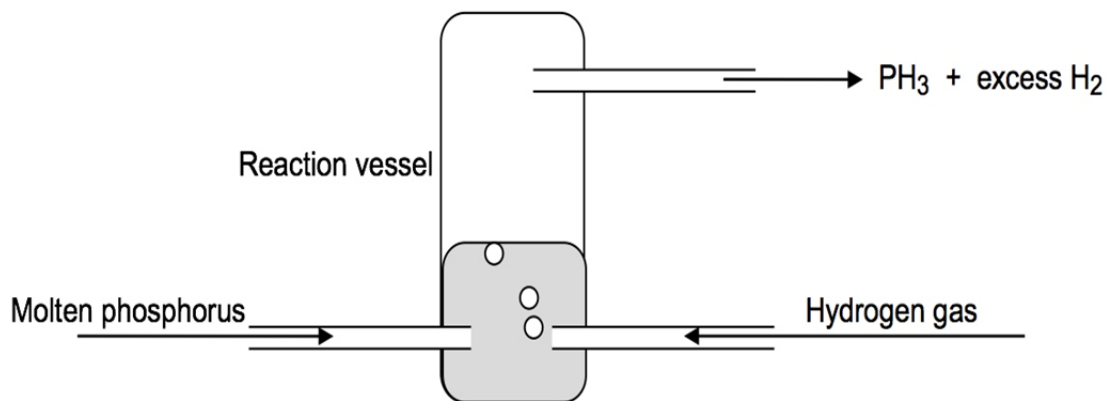
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Question 32**(10 marks)**

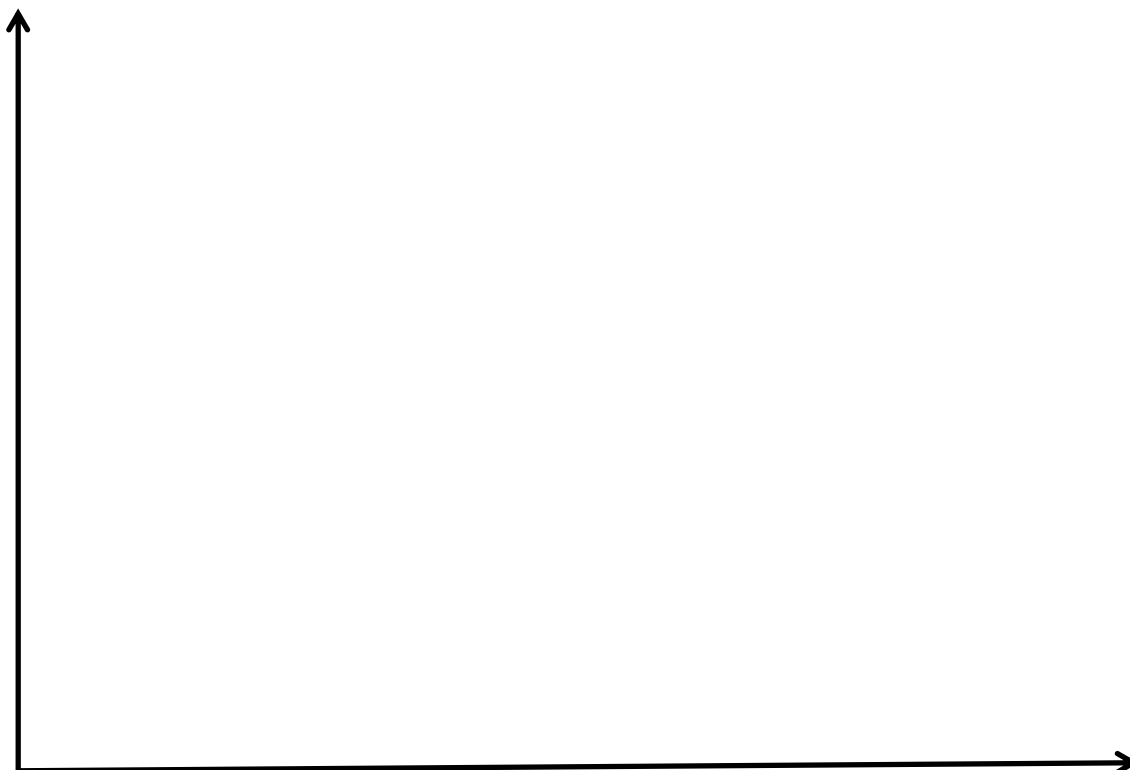
Phosphine (PH_3) is a gas that can be produced by bubbling hydrogen gas through molten phosphorus. The reaction is reversible



Activation energy = 66 kJ



- (a) Use the axes below to draw a labelled energy profile diagram to represent the process. State clearly the axes' headings, reactants, products, activation energy and change in enthalpy (ΔH). (4 marks)



See next page

- (b) Would a high temperature, or a low temperature, **most likely** be used in this industrial process?
(Circle one response below). (3 marks)

Moderate Temperature

Low Temperature

State the effect of temperature on the forward reaction rate and yield of this industrial process.

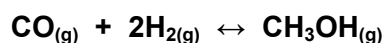
	High Temperature	Low Temperature
Effect on forward reaction rate (write increase, decrease or no change)		
Effect on yield (write increase, decrease or no change)		

- (c) Mixtures of ammonia and phosphine can be separated by distillation due to their different boiling points. Explain why the boiling point of phosphine is lower than that of ammonia even though its molar mass is nearly twice that of ammonia. (3 marks)

	Phosphine (PH ₃)	Ammonia (NH ₃)
Boiling point (°C)	- 88	- 33
Molar mass (gmol ⁻¹)	33.99	17.03

Question 33
(8 marks)

Consider the equilibrium reaction below



$$\Delta H = -90 \text{ kJmol}^{-1}$$

Indicate in the table below whether there would be an increase, decrease, or no change in the concentration of $\text{CO}_{(\text{g})}$ after the changes given in the table are imposed on the system and **equilibrium has been re-established**. Provide a brief explanation for the observation using relevant knowledge of reaction rates and Collision Theory.

Change	Change in concentration of $\text{CO}_{(\text{g})}$ (circle the correct response)	Brief explanation
The volume of the reaction vessel is halved	☐ increase ☐ decrease ☐ no change	
The temperature of the reaction system is doubled	☐ increase ☐ decrease ☐ no change	
Neon gas (Ne) is pumped into the system while keeping the volume constant	☐ increase ☐ decrease ☐ no change	
H_2 gas is pumped into the system while keeping the volume constant	☐ increase ☐ decrease ☐ no change	

See next page

Question 34**(5 marks)**

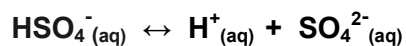
A number of ions can act as an acid or a base in certain conditions. These ions are said to be amphiprotic in nature.

- (a) Hydrogenphosphate (HPO_4^{2-}) is a polyatomic ion that can act as an acid or a base. Use a suitable chemical equation to show that HPO_4^{2-} can act as (2 marks)

i. a **Lowry-Bronsted acid**

ii. a **Lowry-Bronsted base**

- (b) The equilibrium constant (K) for the forward reaction of the equilibrium system below is 1.0×10^{-2} at room temperature.



Is HSO_4^- a strong or weak acid at room temperature? Circle one response below. (1 mark)

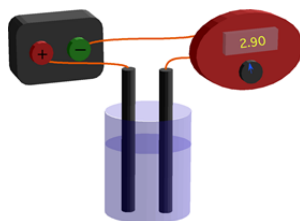
Strong acid

Weak acid

Explain your answer using relevant knowledge of acids and bases theories and equilibrium constant. (2 marks)

Question 35**(10 marks)**

In a Science experiment, a student measures the conductivity (in milliAmps) of three solutions of ethanoic acid. The diagram below shows her experimental set up.



The students' results are listed in the table below

Solution	pH	Electrical current reading (milliAmps)
1	2	20
2	3	9
3	4	4

- (a) **State** the relationship between pH and electrical conducting ability of ethanoic acid for this experiment. (1 mark)

- (b) **Explain** the relationship stated in part (a) using relevant knowledge of solutions and theories of acids and bases. (2 marks)

- (c) Would you expect a 0.1 mol L^{-1} solution of hydrochloric acid to have the same conductivity as that of a 0.1 mol L^{-1} solution of ethanoic acid? Explain your answer using relevant knowledge of the theories of acids and bases. (3 marks)

- (d) Phosphoric acid (H_3PO_4) is a weak triprotic acid.

List all the ions present (in order of decreasing concentration) in a 0.1 mol L^{-1} solution of phosphoric acid. **Explanation is not required.** (2 marks)

- (e) 900.0 mL of water is added to a 100.0 mL solution of a strong acid solution that has a pH of 6. Calculate the pH of the new solution. (2 marks)

Section Three: Extended Answers

40% (80 marks)

This section contains **four (4)** questions. You must answer **all** questions. Write your answers in the spaces provided.

Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression. Lists or dot points are unlikely to gain full marks.

Final answers to calculations should be expressed to **three** significant figures.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

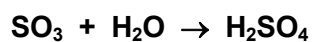
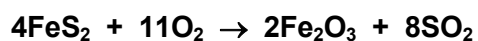
- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

Suggested working time: 80 minutes.

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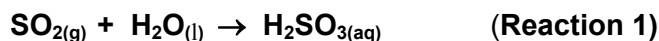
Question 36**(10 marks)**

The preparation of sulfuric acid (H_2SO_4) from iron pyrites (FeS_2) can be represented by the following series of reactions:

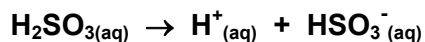


- (a) How many moles of sulfuric acid would be produced from **ONE** mole of iron pyrites? (1 mark)
- (b) What mass of iron pyrites would be needed to produce 123 tonne ($1 \text{ tonne} = 1 \times 10^6 \text{ g}$) of sulfuric acid if the Contact process only converts 95.0% of the sulfur dioxide and the other reactions achieve 100% conversion? (6 marks)

Sulfurous acid (H_2SO_3) can be produced by the dissolving of sulfur dioxide gas in liquid water under the right conditions.



H_2SO_3 is a weak acid that partially ionises to form hydrogen and hydrogen sulfite ions.



- (c) If **Reaction 1** above is slightly **exothermic** then predict the effect of the following changes on the pH of the solution (3 marks)

Change to equilibrium	Effect on pH (write increase, decrease or no change)
A little concentrated nitric acid is added	
Finely powdered $\text{NaOH}_{(s)}$ is added	
The temperature of the system is increased to a new constant value	

Question 37

(20 marks)

Magnesium, a reactive metal found in many common minerals, is also an essential nutrient for both plants and animals.

- (a) Write the electron configuration for magnesium ion (Mg^{2+}). (1 mark)

- (b) Draw the Lewis electron dot diagram for **MgCl_2**

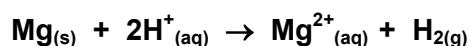
(1 mark)

- (c) Successive ionization energies of magnesium are given in the table below

	First	Second	Third
Energy required / kJmol^{-1}	738	1450	7730

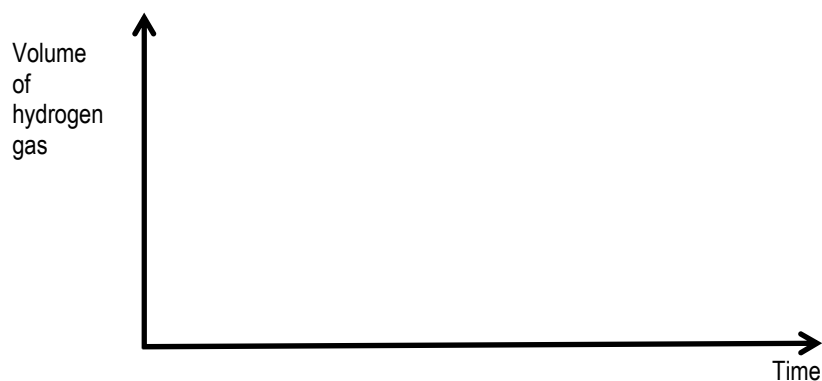
Explain why the third ionization energy is **much** greater than the second ionisation energy. (3 marks)

- (d) Magnesium metal is often used in school laboratories to make hydrogen gas. The equation for the reaction between dilute sulfuric acid and magnesium metal is



In a reaction rate experiment, a student studies the rate of the reaction above by measuring the volume of hydrogen gas produced over time. Assuming no significant errors are made, **sketch** a graph to show what the student is most likely to observe from the volume readings over time.

(1 mark)



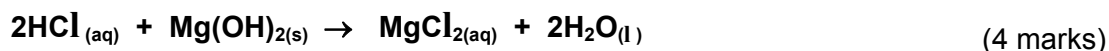
- (e) A well known compound of magnesium is magnesium hydroxide $\text{Mg}(\text{OH})_2$ which is commonly used as an antacid to neutralise stomach acid.

A typical magnesium hydroxide antacid tablet contains 300mg of magnesium hydroxide.

In the stomach, the normal concentration of hydrochloric acid is 0.100 mol L^{-1} . The volume of acid in the stomach can be assumed to be 0.650 litres. Calculate the number of moles of hydrochloric acid in the stomach. (1 mark)

A stressed-out student, studying for her Chemistry exam eats a whole packet of Tim Tams, causing the concentration of the acid in her stomach to rise to 0.200 mol L^{-1} . Calculate the number of moles of hydrochloric acid at this new concentration. (1 mark)

Calculate the number of magnesium hydroxide tablets the student would need to reduce the concentration of acid in her stomach back to the normal level. The non-ionic chemical reaction between stomach acid and magnesium hydroxide is:



- (g) Magnesium hydroxide is only sparingly soluble in water and the equilibrium below exists in a saturated solution.



Use relevant knowledge of reaction rates and Collision Theory to state and explain how changes in pH affect the **solubility** of solid magnesium hydroxide (i.e. more soluble or less soluble) in the equilibrium system above if the temperature is kept constant. (8 marks)

Change: Increase in pH		
Solubility of $\text{Mg}(\text{OH})_2$: (circle one response)	more soluble	less soluble
Explanation: _____		

Change: Decrease in pH		
Solubility of $\text{Mg}(\text{OH})_2$: (circle one response)	more soluble	less soluble
Explanation: _____		

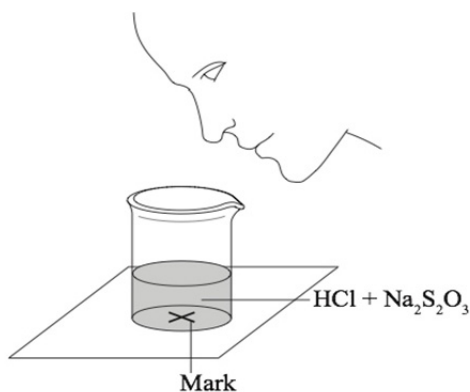
Question 38

(25 marks)

A group of students investigated the rate of reaction between aqueous sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) and hydrochloric acid according to the non-ionic chemical equation below.



The two reagents were rapidly mixed and placed over a mark on a piece of paper. The time taken for the precipitate of sulfur to obscure the mark 'X' (i.e. 'X' can no longer be seen) when viewed through the reaction mixture was recorded.



Initially the students measured out 10.0 mL of 0.500 mol L^{-1} hydrochloric acid and to this solution added 40.0 mL of $0.0200 \text{ mol L}^{-1}$ aqueous sodium thiosulfate. The mark on the paper was obscured 47 seconds after the solution was mixed.

- (a) The teacher asked the students to study the effect of **halving** the concentration of sodium thiosulfate on the rate of the reaction.

i. State the volumes of the liquids that should be mixed.

(2 marks)

Liquid	$0.500 \text{ mol L}^{-1} \text{ HCl}$	$0.0200 \text{ mol L}^{-1} \text{ Na}_2\text{S}_2\text{O}_3$	Water
Volume			

ii. State why it is important that the students use a similar beaker for both occasions.

(1 mark)

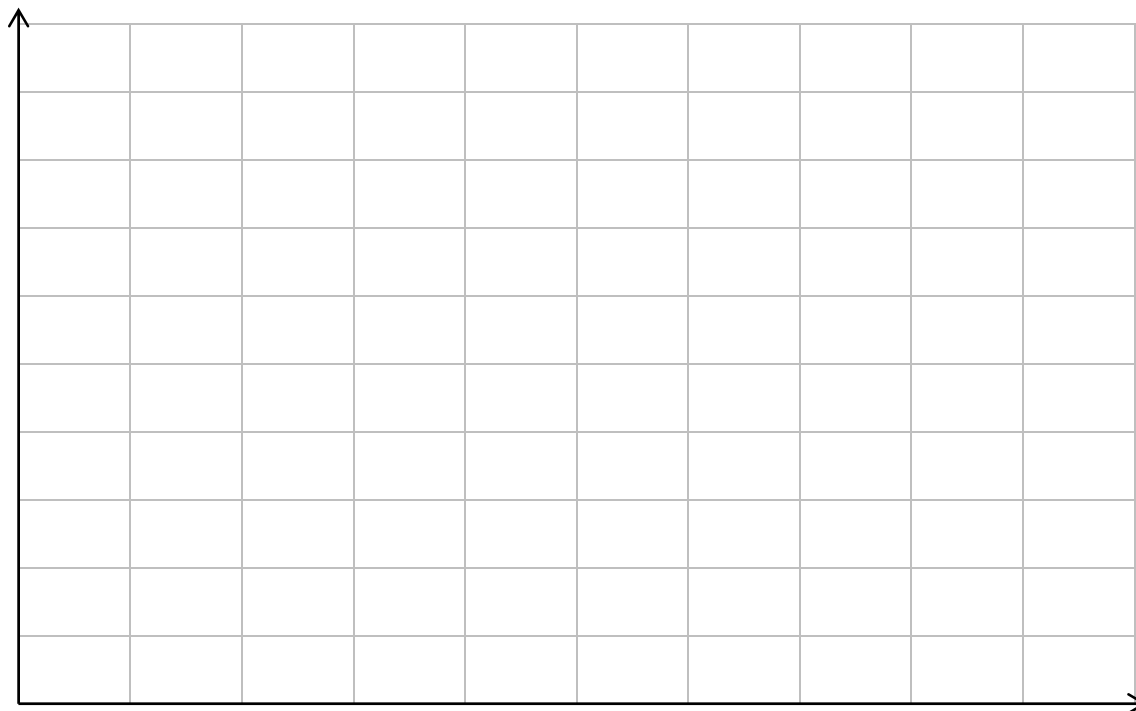
iii. State one (1) safety aspect that was neglected by the instructional diagram above.

(1 mark)

- iv. The results of a **separate** experiment investigating the effect of the concentration of hydrochloric acid of the reaction rate, while keeping the concentration of the sodium thiosulfate at the **original concentration value**, are given in the table below.

Concentration of HCl (molL ⁻¹)	0.020	0.040	0.060	0.080
Time (s)	89.1	72.8	62.4	54.2

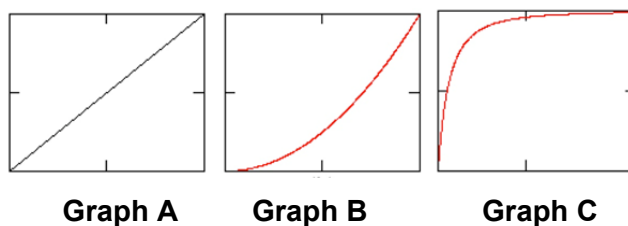
On the axes provided, draw a **fully labelled** graph to plot the information above. (4 marks)



- v. Explain the shape of the graph using relevant knowledge of Collision theory and reaction rates. (2 marks)

- (b) The teacher asked the students to devise another technique to measure the rate of this reaction. One group suggested recording how long it takes for the pH of the solution to change by one unit.

Which of the following graphs **best** represents the change in the pH of the reaction mixture as the experiment proceeds? (Circle one of the graphs below) (1 mark)



See next page

- (c) Another group of students suggested measuring the sulfur dioxide gas evolved over time.
- i. Calculate the volume of sulfur dioxide gas (in litres) that a mixture of 10.0 mL of 0.500 molL⁻¹ hydrochloric acid and 40.0 mL of 0.0200 molL⁻¹ aqueous sodium thiosulfate would produce at 101.3 kPa and 25°C. **Write your final numerical answer in 3 significant figures.** (5 marks)

- ii. Sulfur dioxide gas, a major cause of acid rain, is highly soluble in water. Use your knowledge of relevant bonding theories to explain why this is the case. You may find a suitable diagram useful in your explanation. (2 marks)

iii. The equilibrium equation for the dissolution of sulfur dioxide in water is shown below.



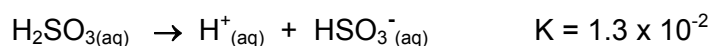
Write an equilibrium expression, K for the reaction above.

(1 mark)

K =

At 0°C, the solubility of sulfur dioxide in water is 228 gL⁻¹ but at a temperature of 90°C, its solubility in water is only 5.8 gL⁻¹. Is the dissolution of sulfur dioxide in water endothermic or exothermic? Explain your answer using relevant knowledge of Collision theory and reaction rates (4 marks)

iv. Sulfur dioxide can form two different acids in water, sulfurous acid (H₂SO₃) and sulfuric acid (H₂SO₄). The ionisation equation and the K value for each acid at 25°C is shown below.



Which of the two acids above is a better electrolyte at 25°C? **Explanation is not required.**

(1 mark)

Write a chemical equation to show that HSO₃⁻ can act as an **Arrhenius acid**.

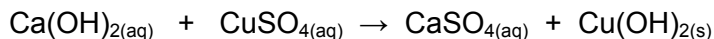
(1 mark)

Question 39**(25 marks)**

Mildews and fungi can cause great damage to grape vines. One chemical spray used to control these diseases is called Bordeaux mixture. A home gardener who wishes to treat his grapes with Bordeaux mixture prepares a mixture using the instructions given below.

- i. Add 25.0 g of calcium hydroxide powder to 25.0 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ powder.
- ii. Mix these powders with a small amount of water to make a paste.
- iii. Add the paste to 5.00 L of water and mix well.
- iv. Use the mixture immediately after preparation.

The non-ionic equation for the reaction that occurs after the powders are mixed with water is



- (a) Determine the limiting reactant for the above reaction. Show all working. (4 marks)

- (b) Calculate the **unused** mass of the excess reactant. (3 marks)

- (c) What colour (if any) will the precipitate have? _____ (1 mark)

Commercially, elemental sulfur (S) can be used to control outbreaks of powdery mildew on grapes.

- (d) Write the electron configuration for sulfide ion? _____ (1 mark)

- (e) Which of the following species would you expect to have a larger radius? (Circle **one** of the answers below).

A SULFUR ATOM

A SULFIDE ION

(1 mark)

See next page

(f) Explain your answer to part (e) using relevant knowledge of atomic structure and bonding. (3 marks)

Sulfur remaining on the grapes after harvest can be converted to a number of undesirable compounds during fermentation in wine production.

A wine chemist uses a method called Atomic Absorption Spectroscopy (AAS) to determine the amount of sulfur remaining on grapes.

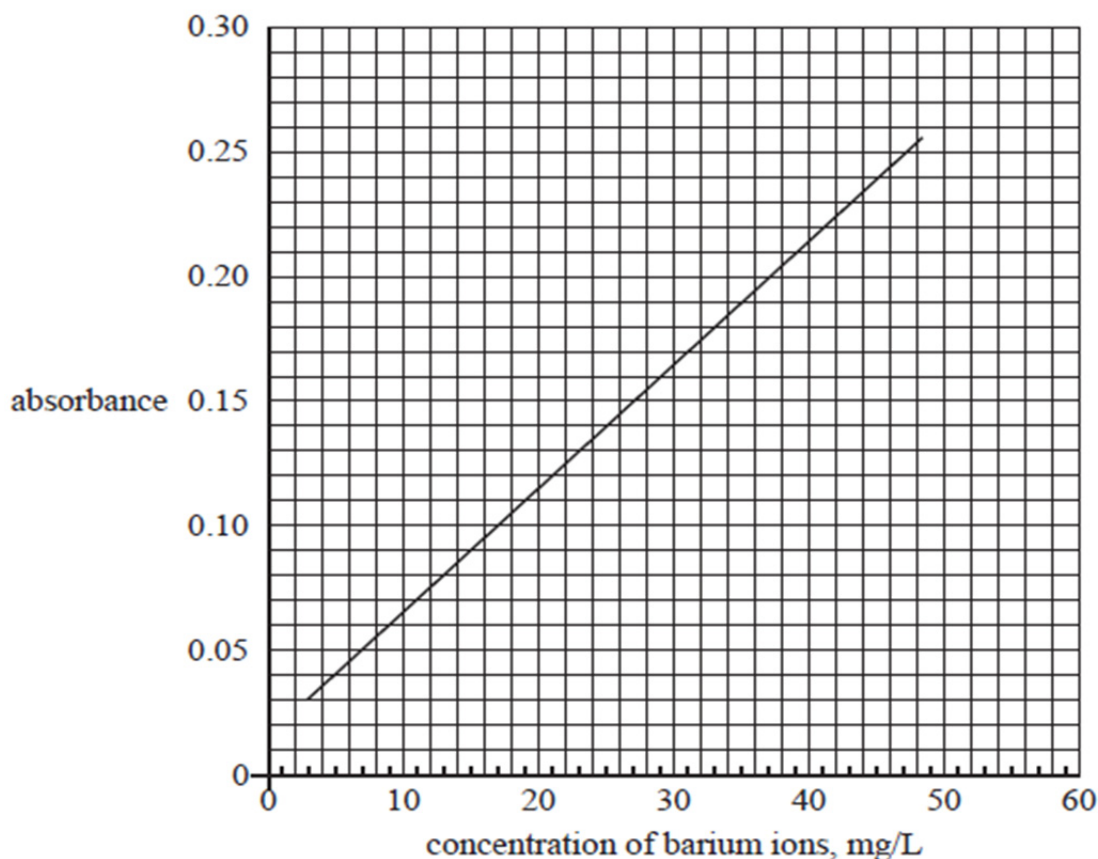
In a particular analysis, 100.0 g of grapes are treated with 100.0 mL of surfactant solution to absorb the sulfur remaining on the grapes when they were harvested.

25.00 mL of this surfactant solution is then treated to convert all of the sulfur to sulfate ions and then dried to produce a powder containing the sulfate ions.

This powder is transferred to a flask containing 2.00 mL of 200 mgL⁻¹ solution of barium Ba²⁺ ions.

The volume of solution in the flask is then made up to 10mL. A precipitate of BaSO₄ forms and settles to the bottom of the flask.

A calibration curve was prepared using standard solutions of 10, 20, 30 and 40 mgL⁻¹ Ba²⁺_(aq).



See next page

- (g) A small amount of the solution containing the unreacted Ba^{2+} ions is removed from the flask and analysed using atomic absorption spectroscopy (AAS). This solution gave an absorbance reading of 0.11.

Use the graph above to determine the concentration of barium ions remaining per 1 L of solution. Hence determine the mass of barium ions, in mg, remaining in **the 10.00 mL sample** solution.

(2 marks)

- (h) Determine the amount of barium ions, in moles, in the original 2.00 mL of 200 mgL^{-1} solution of Ba^{2+} provided and use this value to calculate the amount of barium, in moles, that **reacted** to produce the barium sulfate precipitate.

(3 marks)

- (i) Determine the mass of sulfur, in mg, remaining on the 100 g of harvested grapes.

(4 marks)

One common home remedy for the treatment of mildews involves the use of 'bicarbonate of soda' (potassium or sodium hydrogencarbonate). It is believed that the spraying of this solution makes plant leaves alkaline and thus prevents the germination of mildews' spores.

(j) Write the chemical formula for the **conjugate base** of hydrogencarbonate ion. _____(1 mark)

(k) In a field study, solutions containing 5000 – 10000 ppm of NaHCO_3 were found to be the most effective in controlling mildews growth in but solutions containing over 20000 ppm NaHCO_3 were found to damage and/or severely affect leaf growth in ornamental plants.

Calculate the mass of NaHCO_3 (in grams) required to make 100.0 mL of NaHCO_3 solution with a concentration of 10000 ppm assuming that NaHCO_3 dissolves completely in water and the density of water is 1.0 g mL^{-1} . (2 marks)

END OF EXAM



See next page

Spare answer page

Question number: _____

See next page

Spare answer page

Question number: _____

See next page

Spare answer page

Question number: _____

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